



SEEK WISDOM, ELEVATE YOUR INTELLECT AND SERVE HUMANITY !

Addis Ababa University
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PRETERM LABOUR

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DEFINITION

- A preterm birth is commonly defined as one that occurs after 20⁺ wks & before 37⁺ wks regardless of birthweight.
- ACOG & SMFM have adopted the nomenclature *late preterm & early term* to acknowledge the **contribution of GA** in these ranges **to neonatal risks**.

EPIDEMIOLOGY

- **INCIDENCE**
 - The WHO has estimated ~1 in 10 of all births in 2010 were preterm—almost 15M worldwide.
 - Africa & Asia accounted for almost 11 million

ETIOPATHOGENESIS

- Risk factors for spontaneous preterm birth (sPTB) arise from:
 - Maternal factors,
 - Prior pregnancy hx, and
 - Current pregnancy risks.

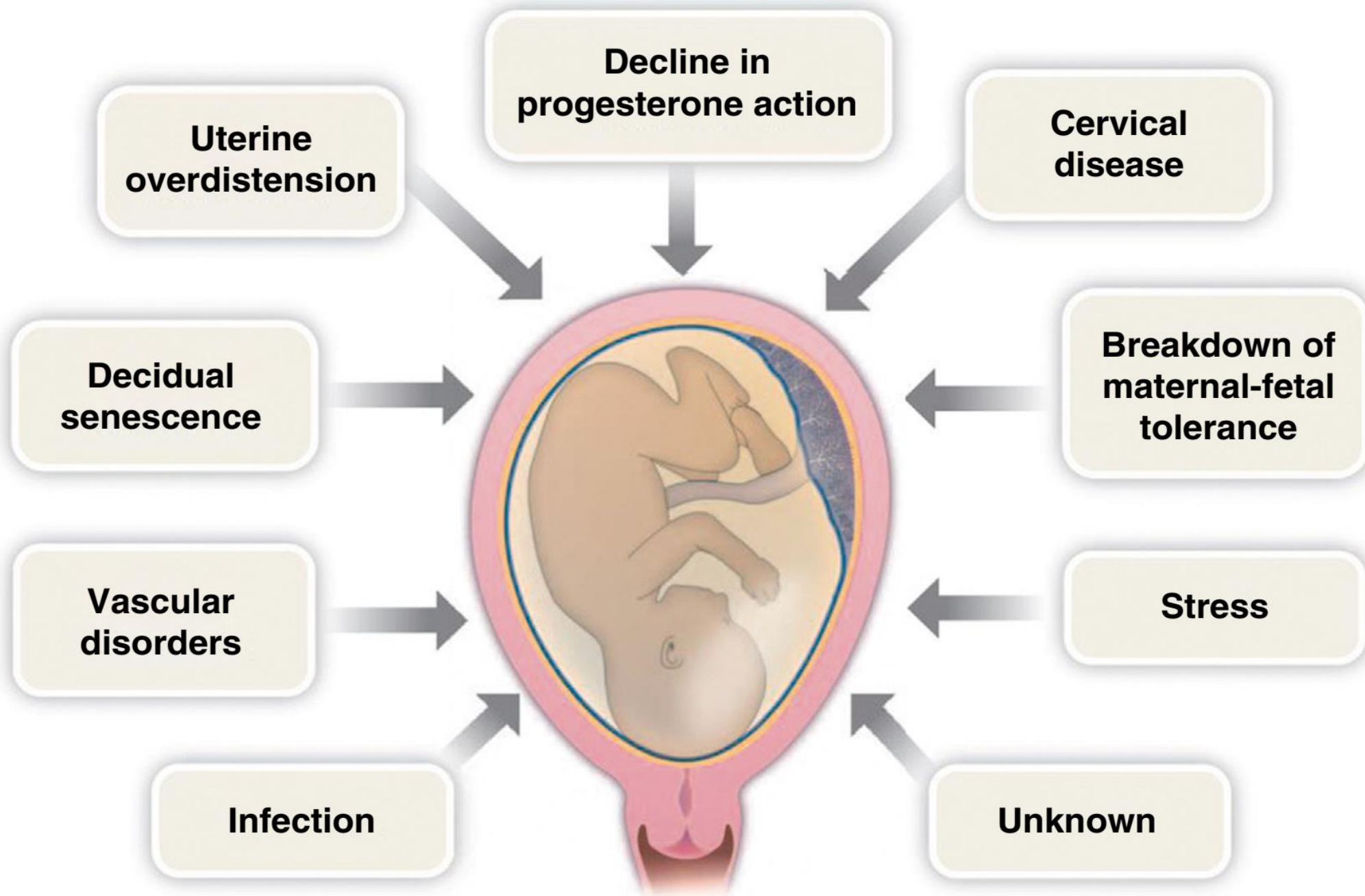


Fig. 36.6 Causes of the Preterm Parturition Syndrome. (From Romero R, Dey SK, Fisher SJ. Preterm labor: one syndrome, many causes. *Science*. 2014;345:760-765.)

THE FOUR MAJOR PATHOGENIC PATHWAYS TO sPTB

#1 Stress-induced premature activation of the
HPA axis

- Causes of stress
- Relationship between stress and hormonal changes leading to sPTB

#2 Inflammation

- Pathophysiology

#3 Decidual hemorrhage

#4 Pathologic uterine distention

- **Final common pathway** - Although each mechanism has distinct epidemiologic, genetic, and clinical characteristics, they are not mutually exclusive. They share a final common pathway involving the formation of uterotonic agents and proteases that weaken the fetal membranes and cervical stroma, eventually leading to preterm labor and birth ( figure 1).

I. Stress-induced premature activation of HPA-axis

- **Causes of stress:**
 - **Maternal stress** (Psychosocial:— eg, depression, posttraumatic stress disorder, anxiety)
 - Can activate the maternal HPA-axis and is associated with <2x **↑d risk of sPTB**.
 - **Fetal stress** can be caused by uteroplacental vascular insufficiency.

Cont'd

DURING PREGNANCY

[**Placental – CRH** secreted by trophoblast, amniochorion & decidua]

Hypothalamic – CRH

- Stimulates pituitary secretion of **ACTH**
- Stimulates adrenal secretion of **cortisol**
- **-ve** feedback
(Inhibits hypothalamic CRH & pituitary ACTH)

① Placental – CRH

- Stimulates Fetal Pituitary–**ACTH** secretion
- Stimulates ↑ fetal Adrenal–**cortisol** secretion,
- Stimulates **placental CRH** release via **+ve** feedback^①

② Placental–CRH

- Stimulates **PG-release** from the amniochorion & decidua
- Stimulate **placental CRH** release via **+ve** feedback^②

- The net effect of these **+ve** & **-ve** feedback loops is:
 - ↑ **CRH** as gestation progresses, with high concentration in the maternal-fetal blood & AF by term, and
 - ↑ **cortisol & PGs** levels.
 - **Prostaglandins** are an integral part of the *final common pathway* leading to parturition
 - They promote **cervical maturation & uterine contractility** ultimately resulting in labor.

Synopsis of hormonal changes:

- ① Stress-induced ↑ in **placental—CRH** production
 - Programs a "**placental clock**" for earlier labor & delivery.

Cont'd

② Placental—CRH

→ Stimulates **PG**-release from the amniochorion & decidua.

→ PGs promote **cervical maturation & uterine contractility** ultimately results in labor

Cont'd

- ③ Placental—CRH **directly** or **indirectly** (via activation of the fetal HPA axis) stimulates fetal adrenal DHEA-S,
- The placenta converts DHEA & DHEA-S to **estrogen**[estr-one (E1), estradiol (E2), & estr-iol (E3)],
 - Which, in the presence of ER- α & $\downarrow$ PR action, increase:
 1. Prostaglandin activity
 2. Gap junction formation,
 3. Oxytocin receptors, and
 4. Myosin light chain kinase & Calmodulin
 - » Enzymes responsible for coordinated muscle contraction.

Cont'd

- ④ Glucocorticoids induce the immunophilin co-chaperone FK506-binding protein 51 (FKBP51), which
 - Binds to the ligand binding site on the **PR & GR**
 - To **inhibit** ligand binding to PR & GR,
 - To **inhibit** nuclear translocation, and receptor-mediated transcription.
 - While this **-ve** feedback circuit prevents pathologic effects accruing from excess glucocorticoids,
 - it also **inhibits PR activity**.
 - Consistent with maturation of the fetal HPA axis & ↑maternal cortisol @term, term—labor is accompanied by ↑d expression of FKBP51 in decidual cells,
 - Resulting in functional progesterone withdrawal.

II. Inflammation

- ① Bacteria may have a **direct role** in the pathogenesis of sPTB.
 - i. Bacterial direct uterotonic effects &
 - Bacteria produce **phospholipase A2**
 - **PG** synthesis & endotoxin,
 - Substances that **stimulate uterine contractions** and can cause preterm labor.
 - ii. Bacterial degradation of membranes
 - Some organisms Produce proteases, collagenases, & elastases
 - That can **degrade the fetal membranes**→Preterm PROM,
 - With subsequent spontaneous or indicated PTB.
- ② Bacteria may have a **indirect role** in the pathogenesis of sPTB.
 - Bacterial induction of maternal-fetal inflammatory responses, leading to uterotonic effects
 - Bacterial ligands bind to TLRs expressed on decidual, amnion, chorion, cervical, placental, & local leukocyte cell membranes.
 - This induces the transcription factor NFκB,
 - **Triggers maternal &/ fetal inflammatory response** in susceptible individuals that is linked to sPTB.

III. Decidual hemorrhage

- Decidual hemorrhage originates in **damaged decidual blood vessels** and presents clinically as:
 - Vaginal bleeding or Retroplacental hematoma formation.
- It is associated with a high risk of **preterm labor & PPROM**.
 - The dev't of **PPROM** in the setting of **abruption** may be related to the high decidual cell expression of **tissue factor**, the primary cellular mediator of hemostasis.
 - Following intrauterine hemorrhage from placental abruption, **decidual tissue factor** combines with **factor-VIIa** to activate **factor-Xa**, which in turn complexes with its **cofactor—Va**, to generate **thrombin**.
 - Thrombin activation has been noted in patients with preterm labor and in asymptomatic patients who subsequently delivered preterm
- Hormonal factors, such as progesterone, play an important modulator role.

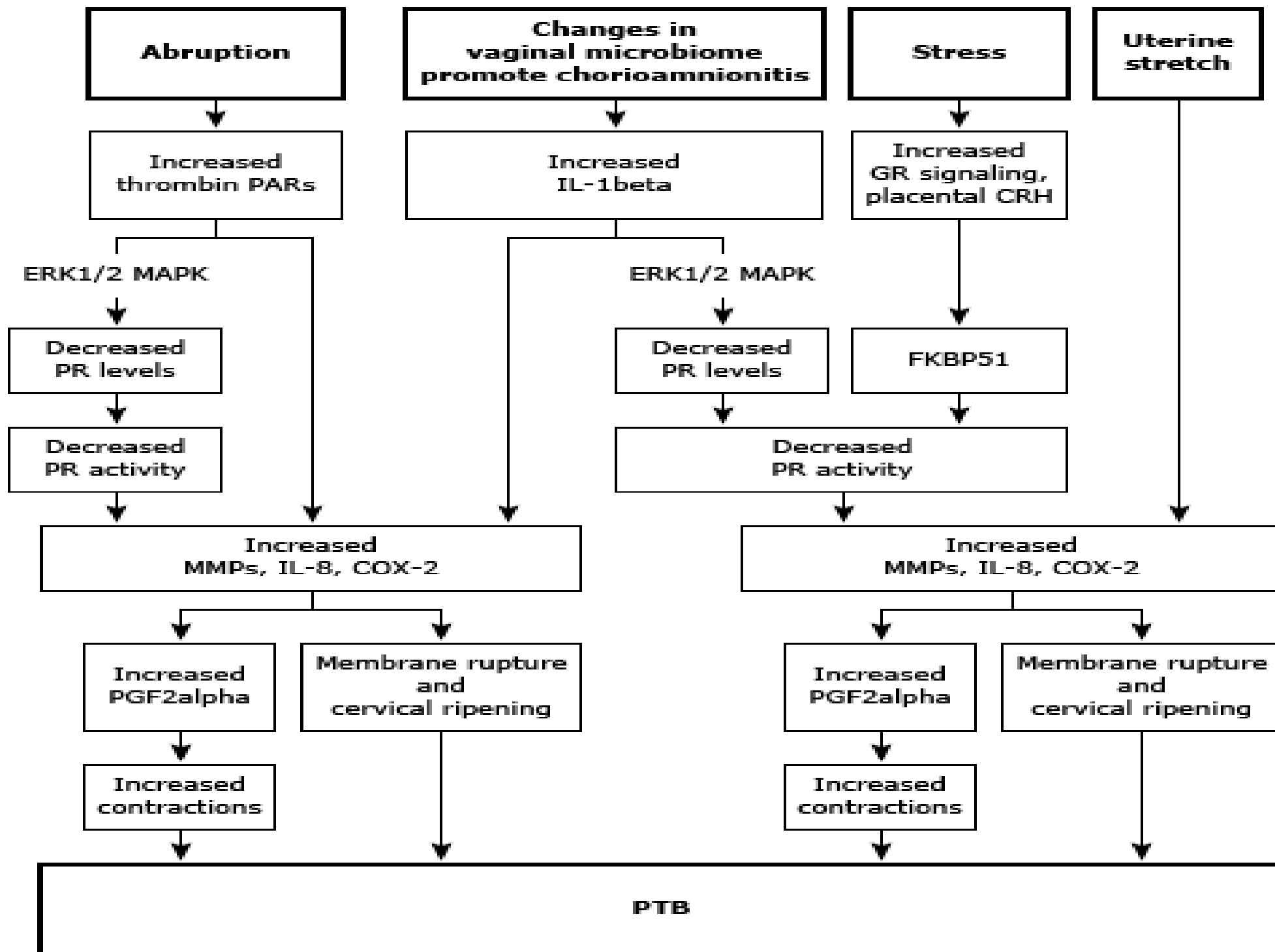
IV. Pathologic uterine distension

- Multiple gestation, polyhydramnios, and other causes of excessive uterine distention are well-described risk factors for sPTB.

① Myometrial distention:

- i. Enhanced **stretching** of the myometrium
 - Induces formation of gap junctions, upregulation of OR, and production of (inflammatory cytokines, PGs, & MLK)
 - Which are critical events preceding uterine contractions & cervical dilation.
- i. ↑expression of genes with important roles in **collagenolysis & inflammation**.

- ### ② Distention of the fetal membranes also contributes to myometrial activation, preterm cervical ripening, and PPRM,
- Likely through the **release of cytokines, prostaglandins, and collagenases**



RISK FACTORS

I. MEDICAL

- Infections
- Periodontal Disease

II. GUT Factors

- Cervical Length
- Cervical Procedures
- Congenital Abnormalities

III. BEHAVIORAL

- Substance Abuse
- Physical Activity
- Nutritional Factors

IV. DEMOGRAPHY, STRESS, & SOCIAL FACTORS

- Black Race

V. GENETIC FACTORS

VI. PREGNANCY HISTORY

VII. CURRENT PREGNANCY RISKS

- APH
- Uterine Overdistention

DIAGNOSIS

- The diagnosis of preterm labor is made clinically based on preterm contractions and cervical changes.
- The presence of risk factors for preterm labor can help establish the diagnosis.
- [Fetal fibronectin](#) levels and cervical length measurements can help assess the risk of impending delivery.
- **INVESTIGATIONS**
 - WBC with differential count
 - Urine analysis/culture and sensitivity
 - Ultrasound (biophysical profile, fetal weight estimation)

MANAGEMENT

- PTL should be managed in a setup where there is best possible neonatal care.
- Management of preterm labour includes:
 - ① **Corticosteroids**
 - Dexamethasone 6 mg IM BID for 48 hrs or betamethasone 12 mg every 24 hrs for 48 hrs.
 - ② **Tocolytics:**
 - **Nifedipine** 20 mg PO loading followed by 10– 20mg Q4–8hrs for up to 48hrs.
 - ③ **Neuroprotection:**
 - Administer MgSO₄ up to 32 weeks of gestation to prevent preterm birth-related neurologic complications.
 - ④ **Antibiotics:** For PTL with unknown GBS status.
 - Ampicillin 2 gm IV as loading dose then 1gm IV Q4hr until delivery for GBS prophylaxis.

CONT'D

WHO—2015 :

- **Tocolysis** (acute & maintenance treatments) are not recommended for women at risk of imminent preterm birth for the purpose of improving newborn outcomes.
- **MgSO₄** is recommended for women at risk of imminent preterm birth **<32wks** of gestation for prevention of cerebral palsy in the newborn.
- **Antibiotic** administration is not recommended for women in **preterm labour** with intact amniotic membranes & no clinical signs of infection.
 - **Erythromycin** is recommended as the Abx of choice for prophylaxis in women with PPROM
 - **Augmentin** is **not recommended** for women with PPROM as it ↑risk of NEC.

PREVENTION OF PRETERM BIRTH

- Address risk factors for preterm labor for all patients, including:
 - Screening for hypertensive pregnancy disorders
 - Screening for gestational diabetes
 - Screening for infections
- For singleton pregnancies:
 - Manage cervical insufficiency and short cervical length, e.g., vaginal progesterone and/or cervical cerclage
- The following measures are not recommended due to a lack of efficacy:
 - Screening for and prophylactic treatment of [bacterial vaginosis](#) in asymptomatic individuals.
 - Prophylactic intramuscular [progesterone](#)
 - Activity restrictions (e.g., bed rest)
- **Screening**
 - Evaluation for [short cervical length](#) can identify individuals at increased risk for preterm birth.

PERINATAL MORTALITY

- Perinatal mortality ↑ markedly as GA and BW decline.
- Major neonatal morbidities related to PTB that carry lifetime consequences include
 - Chronic lung disease,
 - IVH (associated with cerebral palsy),
 - NEC,
 - Visual & Hearing impairment.

REFERENCES

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2. Mark B. Landon, et al. *Gabbe's obstetrics: normal and problem pregnancies*. 8th ed. Philadelphia; Elseivar, 2021.

THANK YOU